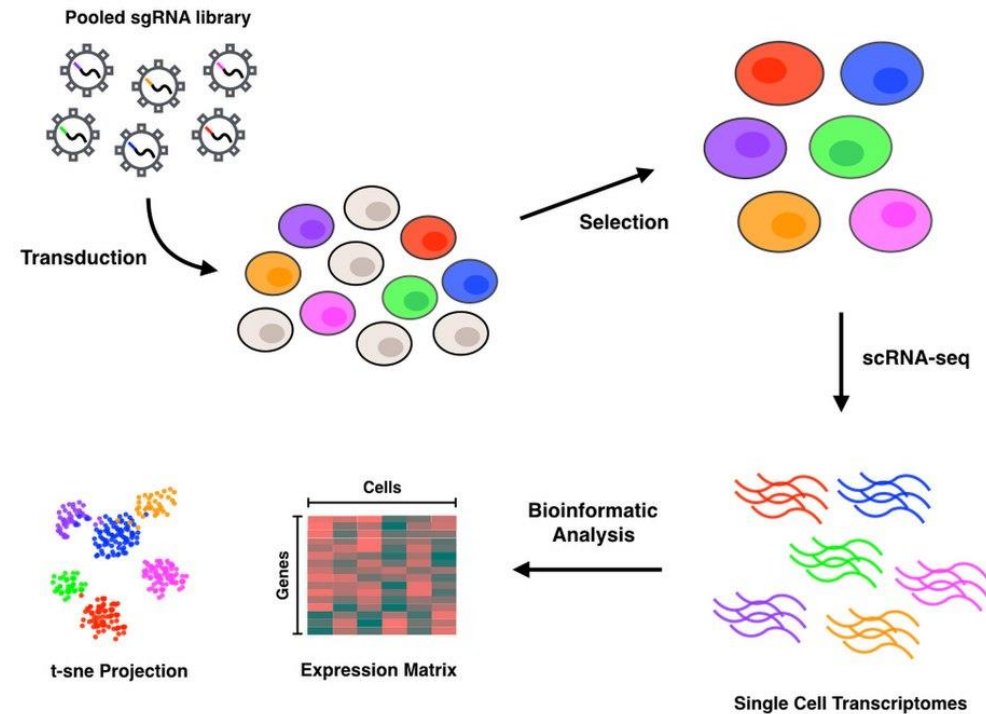


GIDS-AI Biomedical Data Science Hackathon 2025 - Transcriptonaughts

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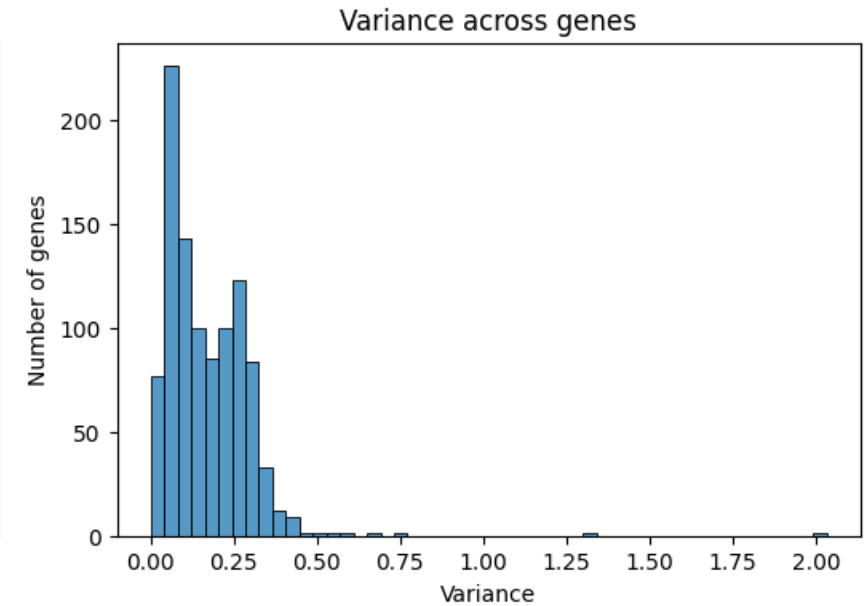
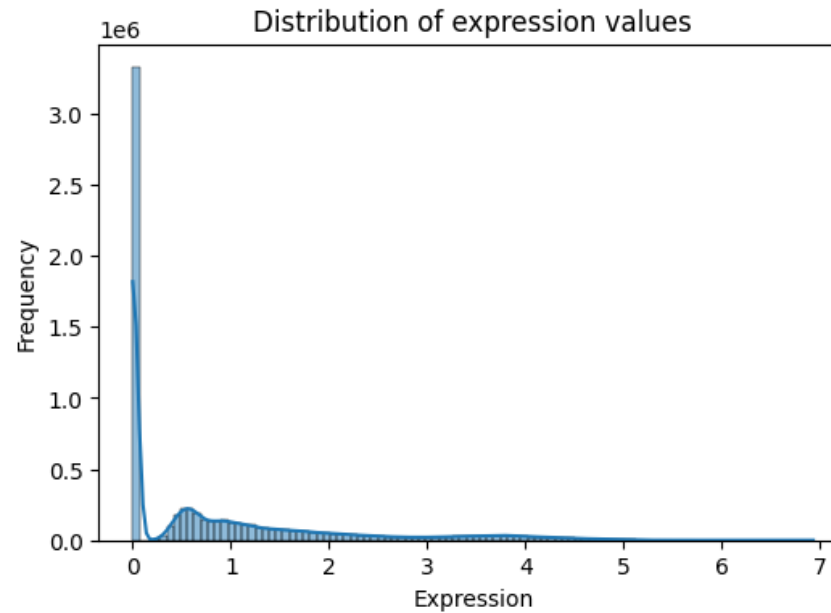
The Challenge

- Predict gene expression when two genes are perturbed (over-expressed) with each other



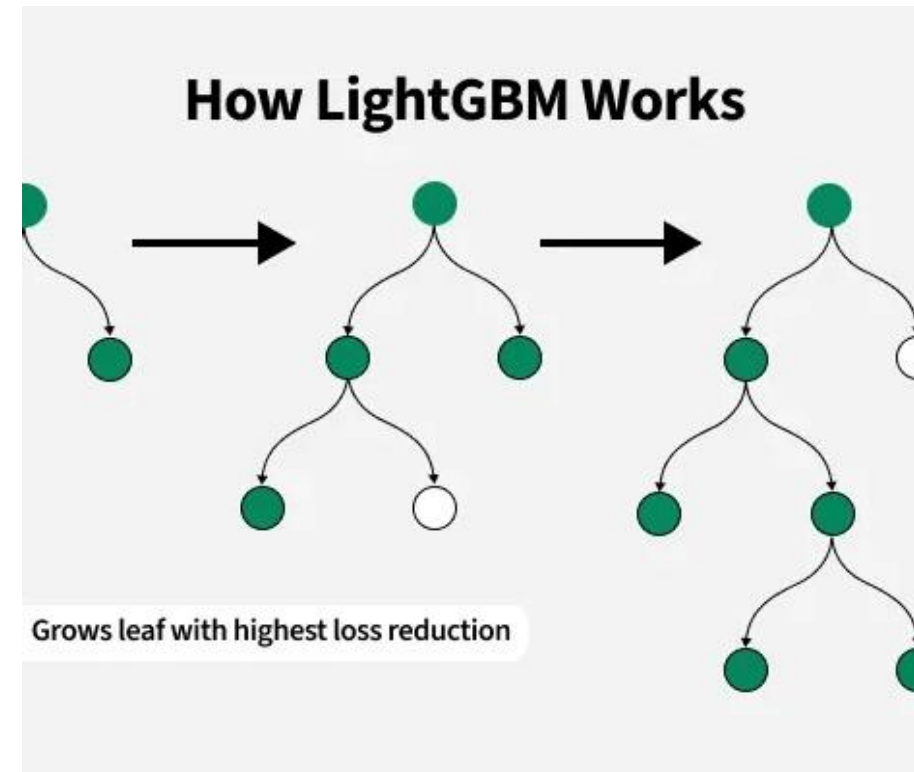
Training Data

- 23.5% of values are 0!
- 4,040 single perturbations (mean 0.861)
- 3,000 double perturbations (mean 0.855)



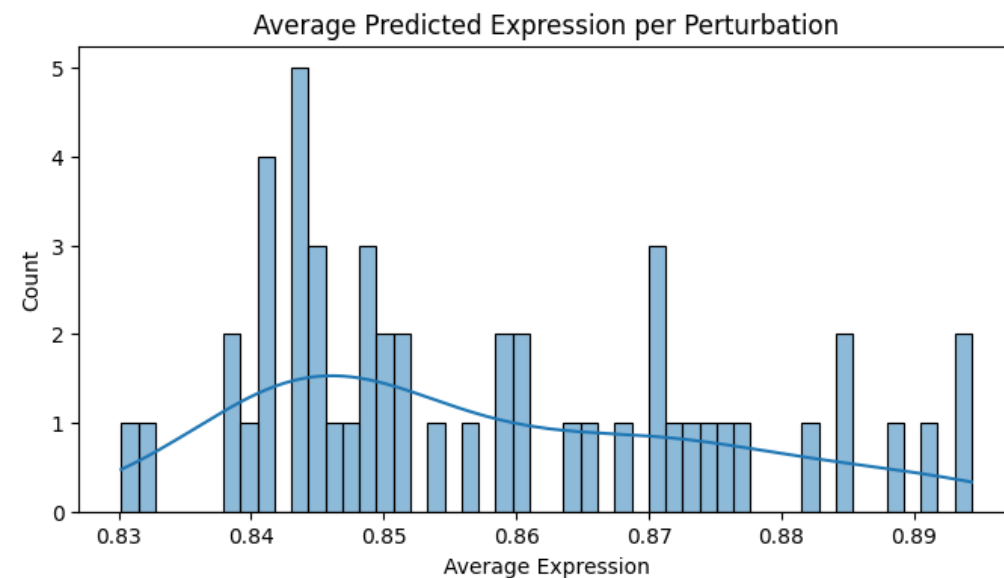
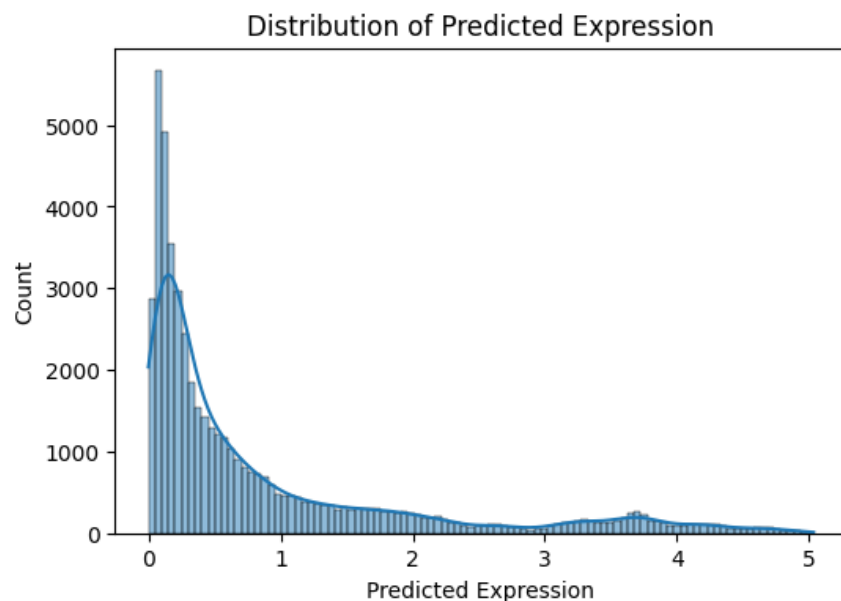
Our Approach

- Preprocess & standardize perturbation profiles, built baseline for missing single-gene data
- Combined single-perturbation profiles & extracted relevant features including product, difference, and average to explore different interaction effects
- Trained one LightGBM regressor per gene to allow each one to have a unique model to predict the double perturbation response
 - LightGBM is much faster/efficient than models such as XGBoost with comparable performance and complexity
- Generated features for unseen data pairs, applied training models and post-processed results (turn all negative values to 0s)



Testing Data Predictions

expression	
count	50000.000000
mean	0.857822
std	1.099642
min	0.000000
25%	0.140235
50%	0.376392
75%	1.075961
max	5.033454



Results & Takeaways

- RMSD score was calculated to 0.14234
- Feature engineering: simple interaction features help capture unique trends/differences in the data
- Modeling: Instead of having one universal model for all genes, training individual ML models per gene helps capture unique expression patterns
- Converting negatives to zero helps reduce error and produce more accuracy to the training data
- Zero-inflation is a huge component of the training data; respecting it instead of removing/clipping zeroes helps better handle these cases

Previous Approach – Two-Stage Model

- Two-stage classifier: predict zero vs. non-zero with LightGBM, Ridge, Elastic regressors
- Over-engineered features with PCA and gene-interaction networks
- Model did not work as well (RMSE = 0.17080) due to abundance of features, over-prediction of zero values, and struggled to model individual gene behaviors
- Our final approach prioritizes simplicity, and treats each gene as an individual unit rather than applying one-size-fits-all

Thank You!

